

Database Normalization Report

National Healthcare Provider Discovery and Appointment Management System

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Submitted On: 11-05-2026

Unnormalized Form (UNF)

Repeating groups (attributes that can appear multiple times for one entity) are shown in braces {}

```
HEALTHCARE_SYSTEM (  
  user_id, first_name, last_name, email, password_hash, doctor_id, patient_id,  
  account_status, license_number, {specializations}, experience_years,  
  consultation_fee, city, bio, is_verified, role_id,  
  phone, address, date_of_birth,  
  hospital_id, hospital_name, hospital_city, hospital_address,  
  appointment_id, appointment_date, appointment_time, appointment_status,  
  review_id, rating, comment,  
  availability_id, day_of_week, start_time, end_time, is_booked,  
  note_id, note_content  
)
```

First Normal Form (1NF)

Issues in Unnormalized Data:

- Repeating specializations

Solution: Create junction tables

```
HEALTHCARE_SYSTEM (  
  user_id, first_name, last_name, email, password_hash, doctor_id, patient_id,  
  account_status, license_number, experience_years, role_id,  
  consultation_fee, city, bio, is_verified,  
  phone, address, date_of_birth,  
  hospital_id, hospital_name, hospital_city, hospital_address,  
  appointment_id, appointment_date, appointment_time, appointment_status,  
  review_id, rating, comment,  
  availability_id, day_of_week, start_time, end_time, is_booked, note_id,  
  note_content  
)  
  
SPECIALIZATIONS (  
  specialization_id, specialization_name  
)  
  
USER_SPECIALIZATION (  
  doctor_id, specialization_id  
)
```

Second Normal Form (2NF)

Functional Dependencies Identified

Determinant	Dependent	Type
user_id	first_name, last_name, email, password_hash, role_id, account_status	Partial – on user_id only
doctor_id	license_number, experience_years, consultation_fee, city, hospital_affiliation, bio, is_verified	Partial – on doctor_id only
patient_id	phone, address, date_of_birth	Partial – on patient_id only
hospital_id	hospital_name, hospital_city, hospital_address	Partial – on hospital_id only
review_id	rating, comment	Partial – on review_id only
appointment_id	appointment_date, appointment_time, appointment_status,	Partial – on appointment_id only
availability_id	day_of_week, start_time, end_time, is_booked, note_id, note_content	Partial – on availability_id only

Decomposition into 2NF Relations

Table	Attributes
USERS	user_id, first_name, last_name, email, password_hash, role_id, account_status
DOCTORS	doctor_id, user_id, license_number, experience_years, consultation_fee, city, hospital_affiliation, bio, is_verified
PATIENTS	patient_id, user_id, phone, address, date_of_birth
ADMINS	admin_id, user_id
APPOINTMENTS	appointment_id, doctor_id, patient_id, appointment_date, appointment_time, appointment_status
REVIEWS	review_id, appointment_id, patient_id, doctor_id, rating, comment
SPECIALIZATIONS	specialization_id, specialization_name
DOCTOR_SPECIALIZATIONS	doctor_id, specialization_id
HOSPITALS	hospital_id, hospital_name, city, address
DOCTOR_HOSPITALS	doctor_id, hospital_id

AVAILABILITY	availability_id, doctor_id, day_of_week, start_time, end_time, is_booked
APPOINTMENT_NOTES	note_id, appointment_id, note_content

Step 4 — Third Normal Form (3NF)

Already in 3NF

Step 5 — Boyce-Codd Normal Form (BCNF)

Already in BCNF